CSE 1141 - COMPUTER PROGRAMMING I

Programming Assignment # 4

DUE DATE: 10/12/2019 - 23:00 (No extension)

In this homework, you will write a program that asks the user to enter two positive integer values: startval, and endval. The program will then find all numbers that fall between startval and endval according to menu that you will prepare.

An Example Run of the program:

Enter the starting value: **25**
Enter the ending value: **150**
Welcome to our Number Finder Program.
1. Prime Numbers between 25 and 150
2. Perfect Numbers between 25 and 150
3. Fibonacci Numbers between 25 and 150
4. Woodall Numbers between 25 and 150
5. Tau Numbers between 25 and 150
6. Smith Numbers between 25 and 150
Please enter your menu choice (0 for exit):

Whenever you choose the menu, the numbers will be listed. If there is no number between the range "No ……. Numbers found between 25 and 150" will be written. At the end of run of your program there will be a question "Do you want to find new numbers (Y/N)" and according to answer the program starts again.
For each of these special numbers you will write different method.
You have to check all inputs those are valid or invalid. If the inputs are invalid, you have to write appropriate messages.

The numbers are listed below:

Prime Numbers

A prime number is a whole number greater than 1 whose only factors are 1 and itself. A factor is a whole numbers that can be divided evenly into another number. The first few prime numbers are 2, 3, 5, 7, 11, 13 and 17.

Perfect Numbers

A perfect number is a number equal to the sum of all its proper divisors (divisors smaller than the number) including 1. The number 6 is the smallest perfect number; because it is the sum of its divisors 1, 2, and 3 (1+2+3 = 6)

Fibonacci Numbers

The Fibonacci Sequence is the series of numbers: 0, 1, 1, 2, 3, 5, 8, 13, 21, 34, ... The next number is found by adding up the two numbers before it.
The Rule is $x_n = x_{n-1} + x_{n-2}$ Where $x_0=0$ and $x_1=1$

Woodall Numbers
A Woodall number is a number of the form $W_n=(2^n.n)-1$.
The first few Woodall numbers are 1, 7, 23, 63, 159 for n=0,1,2,3,4,5

Tau Numbers
Tau number is the number whose number of divisors of n divides n.
As an example 12 is a Tau number.
Divisors of 12: 1,2,3,4,6,12
Number of divisors of 12: 6
12 mod 6 = 0

Smith Number
A Smith number is a composite number the sum of whose digits is the sum of the digits of its prime factors (excluding 1). (The primes are excluded since they trivially satisfy this condition). One example of a Smith number is the beast number 666=2·3·3·37, since 6+6+6=2+3+3+(3+7)=18.

Submission Notes:
1. Please zip and submit your files using filename YourNumberHW4.zip to Canvas system (under Assignments tab).
2. Your zip should contain: Java source code (Pro4_150713852.java) and (Pro4_150713852.class)
3. Write a comment at the beginning of your program to explain the purpose of the program.
4. Write your name and student ID as a comment.
5. Include necessary comments to explain your actions.
6. Select meaningful names for your variables and class name.
7. You are allowed to use the materials that you have learned in lectures & labs.
8. Do not use things that you did not learn in the course.
9. Program submissions should be done through the Canvas class page, under the assignments tab. Do not send program submissions through e-mail. E-mail attachments will not be accepted as valid submissions.
10. You are responsible for making sure you are turning in the right file, and that it is not corrupted in anyway. We will not allow resubmissions if you turn in the wrong file, even if you can prove that you have not modified the file after the deadline.
11. In case of any form of copying and cheating on solutions, all parts will get ZERO grade. You should submit your own work. In case of any forms of cheating or copying, both giver and receiver are equally culpable and suffer equal penalties. All types of plagiarism will result in zero grade from the homework.
12. No late submission will be accepted.

Grading:
- Each of numbers 12 points (6*13=78)
- Taking inputs and necessary controls 15 points
- Submission rules 7 points